

TABLE OF CONTENTS

PREFACE	13
I. BASIC CONCEPTS	15
<i>Introduction, What is programming?, Flowcharting, Information Representation</i>	
II. Z80 HARDWARE ORGANIZATION	46
<i>Introduction, System Architecture, Internal Organization of the Z80, Instruction Formats, Execution of Instructions with the Z80, Hardware Summary</i>	
III. BASIC PROGRAMMING TECHNIQUES	94
<i>Introduction, Arithmetic Programs, BCD Arithmetic Multiplication, Binary Division, Instruction Summary, Subroutines, Summary</i>	
IV. THE Z80 INSTRUCTION SET	154
<i>Introduction, Classes of Instructions, Summary, Individual Descriptions</i>	
V. ADDRESSING TECHNIQUES	438
<i>Introduction, Possible Addressing Modes, Z80 Addressing Modes, Using the Z80 Addressing Modes, Summary</i>	

VI. INPUT/OUTPUT TECHNIQUES **460**

Introduction, Input/output, Parallel Word Transfer, Bit Serial Transfer, Peripheral Summary, Input/Output Scheduling, Summary

VII. INPUT/OUTPUT DEVICES **511**

Introduction, The Standard PIO, The Internal Control Register, Programming a PIO, The Zilog Z80 PIO

VIII. APPLICATION EXAMPLES **520**

Introduction, Clearing a Section of Memory, Polling I/O Devices, Getting Characters In, Testing A Character, Bracket Testing, Parity Generation, Code Conversion: ASCII to BCD, Convert Hex to ASCII, Finding the Largest Element of a Table, Sum of N Elements, A Checksum Computation, Count the Zeroes, Block Transfer, BCD Block Transfer, Compare Two Signed 16-bit Numbers, Bubble-Sort, Summary

IX. DATA STRUCTURES **539**

PART 1—THEORY

Introduction, Pointers, Lists, Searching and Sorting, Section Summary

PART 2—DESIGN EXAMPLES

Introduction, Data Representation for the List, A Simple List, Alphabetic Set, Linked List, Summary

X. PROGRAM DEVELOPMENT **579**

Introduction, Basic Programming Choices, Software Support, The Program Development Sequence, Hardware Alternatives, The Assembler, Conditional Assembly, Summary

XI. CONCLUSION	602
<i>Technological Development, The Next Step</i>	
APPENDIX A	604
<i>Hexadecimal Conversion Table</i>	
APPENDIX B	605
<i>ASCII Conversion Table</i>	
APPENDIX C	606
<i>Relative Branch Tables</i>	
APPENDIX D	607
<i>Decimal to BCD Conversion</i>	
APPENDIX E	608
<i>Z80 Instruction Codes</i>	
APPENDIX F	615
<i>Z80 to 8080 Equivalence</i>	
APPENDIX G	616
<i>8080 to Z80 Equivalence</i>	
INDEX	617